

Quantitative Interpretation of Magnetization Transfer

R. Mark Henkelman, Xuemei Huang, Qing-San Xiang, G. J. Stanisz, Scott D. Swanson, Michael J. Bronskill

Magnetization transfer contrast (MTC) experiments using off-resonance irradiation have been performed with an agar gel model by systematically varying offset frequency, amplitude of the RF irradiation and gel concentration. The experimental results are shown to be quantitatively modelled by a two-pool system consisting of a liquid pool with a Lorentzian line shape and a small semisolid pool with a Gaussian lineshape. The fitted model yields physically realistic fundamental parameters with a T_2 of the semisolid pool of 13 μ s. Further analysis shows that the off-resonance irradiation MTC experiment has significant limitations in its ability to saturate the semisolid pool without directly affecting the liquid component.

Key words: magnetic transfer.

INTRODUCTION

Cross relaxation or spin exchange between water protons and the semisolid component of tissue has recently generated interest in MR imaging under the designation of magnetization transfer (1, 2). Magnetization transfer images have been shown to provide novel contrast which differs from that achieved through conventional (T_1 , T_2 , and T_2^*) relaxation mechanisms. Preliminary clinical studies have shown magnetization transfer contrast (MTC) to have promise in cartilage imaging (3–5) white matter disease (6–8) and lens of the eye (R.S. Balaban, private communication). Further tissue and disease specific contrast are likely to be achieved using MTC as the phenomenon becomes better understood and is characterized in a variety of tissues.

One immediate and practical application of MTC has been in the suppression of background signal in MRA (9). Short self-refocusing polynomial, Gaussian or hard pulses (10–12) leave the slowly relaxing liquid component of tissue and blood unaffected but saturate spins in the semisolid component due to their very short T_2 . This technique reduces signal intensities of white matter to 51%, grey matter to 61%, and muscle to 55%, while CSF and blood are only attenuated to 96% and 83% (9). Combined with conventional techniques for static spin suppression in MRA, the additional attenuation provided by

MTC yields dramatically improved maximum intensity projection angiograms, particularly in terms of small vessel visibility. The lack of a quantitative underlying model of the magnetization transfer process, however, makes it difficult to ensure that optimum improvement has been achieved (13).

It has also been recognized that MTC can play an unanticipated and incidental role in MR imaging when many RF pulses are applied in short succession (14). Some of the anomalous contrast seen in fast spin echo (FSE) images arises from MTC effects (15). Such findings will be increasingly prevalent with faster and more RF intensive pulse sequences. Avoidance or interpretation of unanticipated MTC effects also requires quantitative models of the underlying mechanisms.

Finally, it is increasingly appreciated that cross relaxation makes an important contribution to tissue relaxation (16–20). Thus, to investigators (such as ourselves) who are interested in a detailed understanding of tissue relaxation, magnetization transfer experiments provide an additional dimension to tissue relaxation analysis. To be useful for probing tissue relaxation, magnetization transfer must first be understood quantitatively in its own right.

It is the purpose of this paper to analyze a comprehensive set of magnetization transfer experimental data from a simple model system consisting of agar gel. A two-pool model consisting of a liquid component with a Lorentzian line shape and a semisolid component with a Gaussian line shape is shown to describe this experimental system accurately. The underlying physical parameters which can (and those which cannot) be elucidated from such experiments are determined and are shown to be consistent with other relaxation measurements. Field dependence of the magnetization transfer is explored. Finally, the parameters of the model system are used to investigate the limitations of simple, off-resonance irradiation techniques for MTC.

METHOD

Agar gel (Signa, St. Louis) samples of 2%, 4%, and 8% by weight were prepared in distilled water (21). An additional sample of 0.3 mM MnCl_2 was studied as a control. NMR measurements were made on a 30-cm, horizontal bore superconducting magnet (Nalorac, Martinez, CA) operating at 1.5 T equipped with a spectroscopy console (SMIS, Surrey, England). All experiments were also repeated on the same equipment at 0.6 T. NMR signal intensities were measured using the pulse sequence shown in Fig. 1 and were normalized to 1.0 for no off-resonance irradiation. Reproducibility of normalized magnetization signals was better than 0.5%. The effect of any residual

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From the Sunnybrook Health Science Centre and Departments of Medical Biophysics and Radiology, University of Toronto, Toronto, Ontario, Canada; and the Department of Radiology, University of Michigan, Ann Arbor, Michigan (S.D.S.).

Address correspondence to: R. Mark Henkelman, Ph.D., Medical Physics Research - RmS132, Sunnybrook Health Science Centre, 2075 Bayview Avenue, Toronto, Ontario, M4N 3M5 Canada.

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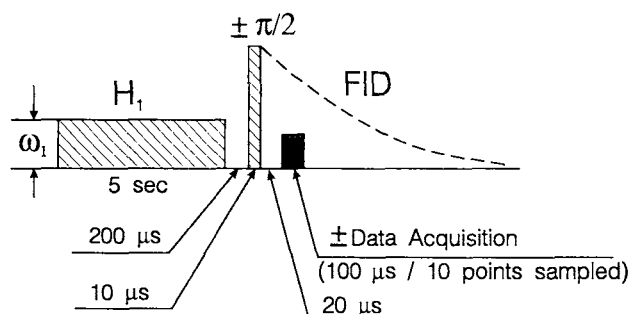


FIG. 1. The pulse sequence for the off-resonance irradiation experiment. Any residual transverse magnetization following the off-resonance RF was compensated by phase cycling both the $\pi/2$ excitation and the data acquisition. Ten 16-bit data points were acquired at a rate of 10 μ s per point and averaged to improved signal-to-noise. The T_2^* of the FID was ≥ 12 ms.

transverse magnetization following the off-resonance irradiation was removed by phase cycling the sign of both the $\pi/2$ pulse and the data acquisition. Off-resonance irradiation frequencies (Δ) were uniformly distributed along a logarithmic scale at 26 points from 0.014 to 213 kHz. The H_1 amplitude was increased in five steps by factors of 2 from 0.17 to 5.34 kHz and was calibrated by determining the length of a rectangular pulse required to invert the spin population.

The duration of the off-resonance irradiation pulse was 5 s to ensure steady state had been reached. Fig. 2 shows the approach to steady state and the measured $T_{1\text{sat}}$ values for two different offset frequencies confirming that 5 s corresponds to more than 10 time constants and is, therefore, adequate to achieve equilibrium. The time between data acquisition and the start of the next saturation pulse was set to 3 s. Thus, measurements of one sample, with 26 offset frequencies, 6 H_1 amplitudes, and 2 phase cycles took a total time of about 50 min. During this time, the RF heating of the sample remained less than 0.3°C.

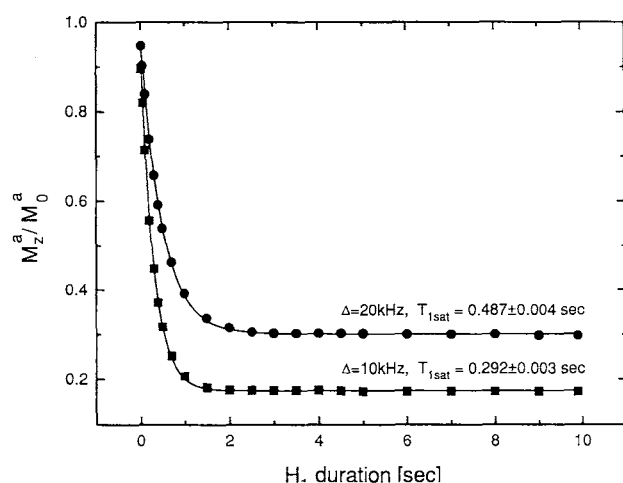


FIG. 2. The approach of the normalized residual longitudinal magnetization to steady state as a function of the duration of the RF pulse for the 2% agar sample and an H_1 amplitude of 0.67 kHz. Durations of 5 s are sufficient to ensure saturation.

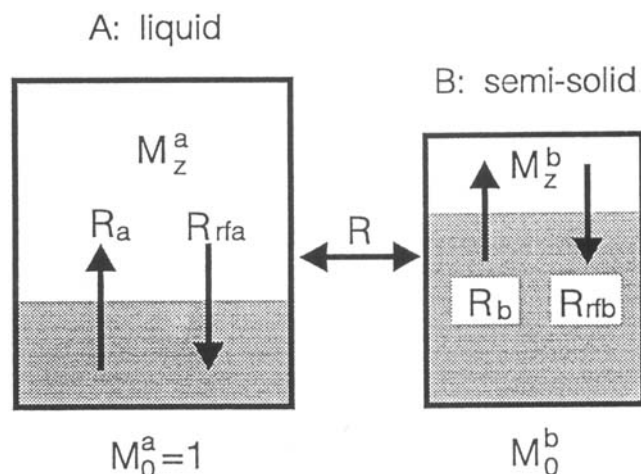


FIG. 3. The model of the agar gel system. The shaded region in each pool represents spins which are not aligned longitudinally. All rates and spin populations are defined in the text and are incorporated in Eq. [9].

THEORY

The model used to analyze the experiments is shown in Fig. 3. It consists of two pools. Pool A is liquid and consists of longitudinal magnetization (unshaded) and other magnetization (shaded) which may be transverse or saturated. Pool B corresponds to the semisolid protons in the sample and is divided similarly into longitudinal and other magnetization. In each of the pools, there is a rate (R_a or R_b) corresponding to T_1 which governs the recovery of longitudinal magnetization. The rate of loss of longitudinal magnetization is R_{RF} , due to the off-resonance irradiation. The exact form of this rate will be discussed later. The total number of spins in pool A is M_0^a which is constant and set equal to 1.0 to normalize the experiment.

The exchange rate between the pools is characterized by a fundamental rate constant, R . Because the pools are in steady state, R is modified by the relative populations of the pools to produce pseudo-first-order rate constants RM_0^b (exchange $A \rightarrow B$) and RM_0^a (exchange $B \rightarrow A$). Thus, the rate of exchange in either direction between these two pools is $RM_0^a M_0^b \text{ s}^{-1}$. In this parameter, we have differed from the work of others (22, 23) by keeping the model symmetrical with respect to the two pools and by using a rate constant as a fundamental parameter, independent of the concentration of gel. This approach could apply equally well to chemical exchange; there is no difference in the mathematics (24).

The Bloch equations for this coupled system can be written as follows:

$$\frac{dM_z^a}{dt} = R_a(M_0^a - M_z^a) - RM_0^b M_z^a + RM_0^a M_z^b + \omega_1 M_y^a \quad [1]$$

$$\frac{dM_z^b}{dt} = R_b(M_0^b - M_z^b) - RM_0^a M_z^b + RM_0^b M_z^a + \omega_1 M_y^b \quad [2]$$

$$\frac{dM_x^{a,b}}{dt} = -\frac{M_x^{a,b}}{T_{2a,b}} - 2\pi\Delta M_y^{a,b} \quad [3, 4]$$

$$\frac{dM_y^{a,b}}{dt} = -\frac{M_y^{a,b}}{T_{2a,b}} + 2\pi\Delta M_x^{a,b} - \omega_1 M_z^{a,b} \quad [5, 6]$$

where $M_{x,y,z}^{a,b}$ are the x, y, and z components of the magnetization of pools A and B respectively. ω_1 is the angular frequency of precession induced by the off-resonance saturating RF pulse and is a measure of the amplitude of the H_1 field. Δ is the frequency offset of the H_1 field in Hz. $R_{a,b}$ are the longitudinal rate constants and $T_{2a,b}$ are the transverse relaxation times for pools A and B, respectively.

For a steady state to exist, all six time derivatives in Eq. [1–6] must be equal to zero which allows solution for the transverse components (x, y) of the magnetizations and produces coupled equations for the longitudinal components.

$$0 = R_a(M_o^a - M_z^a) - RM_o^b M_z^a + RM_o^a M_z^b - \frac{\omega_1^2 T_{2a}}{1 + (2\pi\Delta T_{2a})^2} M_z^a \quad [7]$$

$$0 = R_b(M_o^b - M_z^b) - RM_o^a M_z^b + RM_o^b M_z^a - \frac{\omega_1^2 T_{2b}}{1 + (2\pi\Delta T_{2b})^2} M_z^b \quad [8]$$

Solving for M_z^a by eliminating M_z^b and recalling that M_o^a is normalized to 1:

$$M_z^a = \frac{R_b RM_o^b + R_{rfa} R_a + R_b R_a + R_a R}{(R_a + R_{rfa} + RM_o^b)(R_b + R_{rfb} + R) - RRM_o^b} \quad [9]$$

where:

$$R_{rfa} = \frac{\omega_1^2 T_{2a}}{1 + (2\pi\Delta T_{2a})^2}, R_{rfb} = \frac{\omega_1^2 T_{2b}}{1 + (2\pi\Delta T_{2b})^2} \quad [10, 11]$$

With the exception of the exchange rate which has been discussed previously, these equations are identical to those of Caines *et al.* (22) and Wu (23). Transverse spin exchange is included in the coupled equations as part of the nonlongitudinally aligned component. However, any net transverse magnetization exchange is not modelled explicitly, since the very short T_2 relation time of pool B will destroy any transverse coherence between the two pools (25–27).

The RF rates have been introduced as algebraic summaries in Eq. [9] and are proportional to the Lorentzian absorption line shape of each pool. This description applies only in the steady-state limit and is not formally correct in the transient approach to steady state. The initial oscillations in longitudinal magnetization which occur immediately following the application of off-resonance irradiation and are part of the exact solution of the coupled Bloch equations (28), are not described by RF rates. Thus, this approach should not be used without ensuring that the assumptions are correct in calculations of the transient approach to steady state. These transients

are not of concern if the RF is off.

As shown later, a Lorentzian line shape is not adequate for the semisolid component. As originally suggested by Swanson (29), we have substituted alternative line shapes in the expressions for R_{rfa} . In particular, we have used a Gaussian line shape:

$$R_{rfa} = \omega_1^2 \sqrt{\frac{\pi}{2}} T_{2b} e^{-\frac{(2\pi\Delta T_{2b})^2}{2}} \quad [12]$$

There is, thus, no longer a solution of the coupled Bloch equations, because a single T_2 rate cannot give rise to anything other than a Lorentzian line shape. Gaussian line shapes arise in NMR of solids from strong dipolar couplings among multiple spins with constrained geometrical relationships (30, 31). We have been unsuccessful in modifying the Bloch equations (or for that matter, the density matrix formalism) to yield Gaussian line-shapes.

As has been observed by others (22), unique sets of model parameters (R_a , T_{2a} , R_b , T_{2b} , R , M_o^b) cannot be obtained by fitting Eq. [9] to experimental data. This is to be expected since experiments performed in the “steady state” contain no explicit reference to absolute time scales, but are informative only about relative times and rates. By noting that $(2\pi\Delta T_{2a})^2 \gg 1$ for all significantly non zero points in this experiment, Eq. [9] can be simplified to only five parameters as:

$$M_z^a = \frac{R_b \left[\frac{RM_o^b}{R_a} \right] + R_{rfa} + R_b + R}{\left[\frac{RM_o^b}{R_a} \right] (R_b + R_{rfb}) + \left(1 + \left(\frac{\omega_1}{2\pi\Delta} \right)^2 \left[\frac{1}{R_a T_{2a}} \right] \right) (R_b + R_{rfb} + R)} \quad [13]$$

This reduced equation was fitted to the experimental results using a least squares technique (32). Five model parameters (R_b , T_{2b} , R , $[RM_o^b/R_a]$ and $[1/R_a T_{2a}]$) were uniquely determined although in two cases only ratios of the fundamental physical rate constants to R_a could be obtained. Errors for the fitted parameters were taken as square roots of the diagonal elements of the covariance matrix.

RESULTS

Experimental measurements for the $MnCl_2$ solution at 1.5 T are shown in Fig. 4. For an aqueous paramagnetic solution, there is no semisolid pool and the appropriate model includes only pool A ($M_o^b = 0$, $R = 0$). The solid lines are not fitted to the data but are calculated directly using measured relaxation times of $T_1 = 431$ ms and $T_2 = 45$ ms. Agreement with the experimental measurements is excellent. These control data are shown because they do *not* represent magnetization transfer but only the direct or “bleed over” effect (1) of the off-resonance irradiation on the liquid pool. The sigmoidal shape of the curves simply reflects the Lorentzian line shape plotted

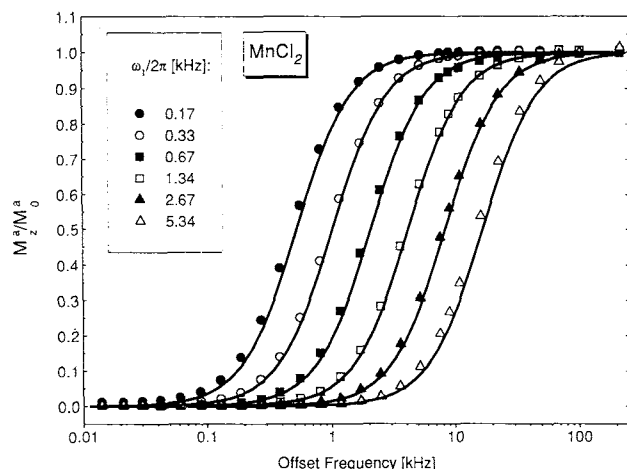


FIG. 4. Magnetization transfer experiments for an aqueous solution of 0.3 mM MnCl_2 . Normalized residual longitudinal magnetization (M_z^a/M_0^a) is plotted versus the frequency offset (Δ) on a logarithmic scale. Data for six different amplitudes of H_1 are shown. The solid lines are calculated from independent measurements of the T_1 and T_2 of the solution.

as a function of the logarithm of resonance offset. We plot all our data in this form because it gives a simple visualization of the lineshape and because it allows the direct irradiation effect to be appreciated immediately. When $(2\pi\Delta T_2)^2 \gg 1$, the offset frequency at half maximum saturation increases linearly with the H_1 amplitude and is given by:

$$\Delta_{\frac{1}{2} \max} = \frac{\omega_1}{2\pi} \sqrt{\frac{T_1}{T_2}} \quad [14]$$

Equation [14] shows that it would be easy to choose an RF amplitude for two doped water solutions with quite different T_1/T_2 ratios such that one solution would be unaffected and the other would be completely saturated. Imaging experiments where the signal intensity of one tissue reduces and that of another does not, need not involve magnetization transfer and may only reflect relative tissue relaxation ratios.

Data for the 2% agar gel at 1.5 T are shown in Fig. 5. (The data for the 4% and 8% agar are shown completely, along with the 2%, in Fig. 7). The solid line in Fig. 5 is the fit of Eq. [13] to the data with a Gaussian line shape for pool B and agreement is excellent. The average residual deviation per point between experiment and theory was 0.015. Similar fits were obtained for the 4% and 8% agar concentrations and the residual deviations were 0.015 and 0.017, respectively. For comparison, a fit to a model which assumes that the semisolid pool has a Lorentzian lineshape is shown in Fig. 6 for the 2% agar. The residual deviation per point is 0.083 and the model deviates significantly and systematically from the measurements. This fit is not acceptable. Thus, calculations of MTC effects based only on the Bloch equations would be incorrect for this agar experimental system.

Reliable fitting, such as Fig. 5, can only be achieved if multiple RF amplitudes are used. Even with the reduced number of parameters of Eq. [13], the fitting of a single RF amplitude does not converge to stable parameters. Thus,

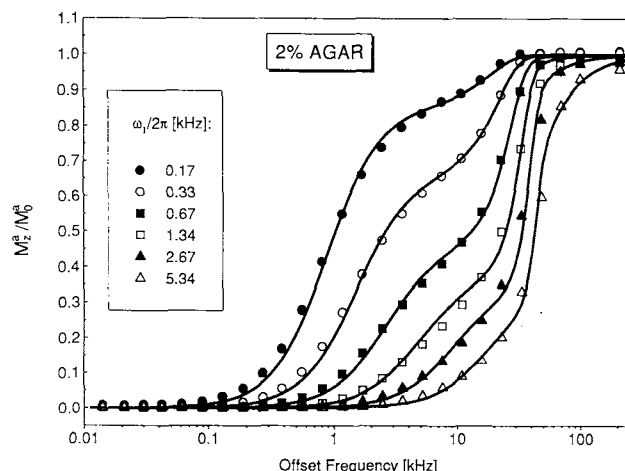


FIG. 5. Magnetization transfer experiments for a 2% by weight agar gel. Data are shown for six different RF amplitudes and 26 different offset frequencies. Solid lines are a fit of Eq. [13] to the data. The average residual deviation per data point is 0.015.

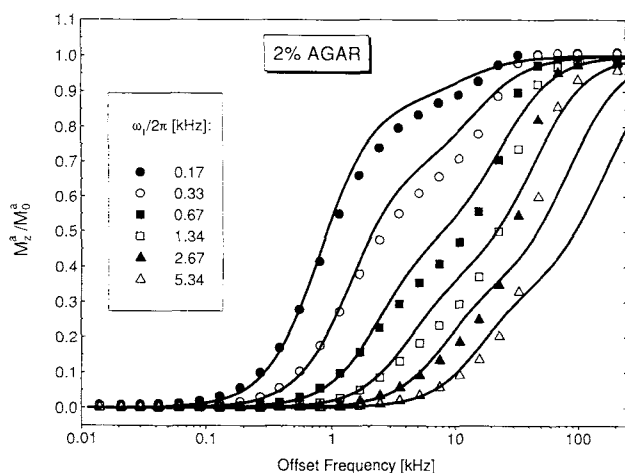


FIG. 6. The same data as in Fig. 5 fitted by Eq. [13] with a Lorentzian lineshape for the semisolid pool. The residual deviation per point is 0.083 and the fit is not acceptable.

for meaningful characterization of the underlying rate constants of the model, fitting to comprehensive measurements over a range of frequency offsets and RF amplitudes is required. To achieve a global fit of the complete data set of 468 measurements, we require additional relationships between the measurements made at different gel concentrations. Specifically, the relaxation rates of the liquid pool are expected to increase linearly with gel concentration in % (C).

$$R_a(C) = R_w + S_1 C \quad [15]$$

$$\frac{1}{T_{2a}}(C) = R_w + S_2 C \quad [16]$$

where R_w is the longitudinal or transverse relaxation rate of pure water (assumed to be equivalent) and S_1 and S_2 are the longitudinal and transverse relaxivities for gel without magnetization transfer in $\text{s}^{-1}/\%$ agar. This assumes that, over and above magnetization transfer, agar

provides other relaxation mechanisms for water protons such as rotational hindrance at surfaces. For the global fit, it is also assumed that the size of pool B is proportional to the concentration of gel. Since the concentration of B is normalized to the concentration of pool A which is defined to be 1,

$$M_o^b = k \frac{C}{100\% - C} \quad [17]$$

where k is some proportionality constant to be determined. All the other parameters remain invariant as the agar concentration is varied and the global fit is carried out with six free parameters (T_{2b} , R_b , R , R_w/k , S_1/R_w , S_2/R_w).

The global fit to the data is shown in Fig. 7 (A–C) for the 2%, 4%, and 8% agar samples. The average residual deviation per point between experiment and theory for all three concentrations is 0.017. The resultant parameters are shown with error estimates in Table 1.

The actual physical parameters of the model cannot be determined without independent information about R_a , the longitudinal relaxation rate of the liquid pool without the interference of pool B. The longitudinal relaxation rate (R_a^{obs}) measured from a simple inversion recovery experiment is inappropriate, however, because it includes the effect of coupling to pool B even if pool B is not being irradiated. Fortunately, the effect of pool B on the measured longitudinal relaxation rate can be dealt with explicitly. If we start with the coupled longitudinal relaxation Eq. [1 and 2] *without* off-resonance irradiation, they can be solved in the transient regime to give relaxation rates as follows (25, 33):

$$\frac{dM_z^a}{dt} = R_a(M_o^a - M_z^a) - RM_o^b M_z^a + RM_o^a M_z^b \quad [18]$$

$$\frac{dM_z^b}{dt} = R_b(M_o^b - M_z^b) - RM_o^b M_z^b + RM_o^b M_z^a \quad [19]$$

Solving for $M_z^a(t)$

$$M_z^a(t) = C_1 e^{-\lambda_1 t} + C_2 e^{-\lambda_2 t} + M_o^a \quad [20]$$

where

$$\lambda_{1,2} = \frac{1}{2} \left([RM_o^b] + R_a + R + R_b \pm \sqrt{([RM_o^b] + R_a - R - R_b)^2 + 4R[RM_o^b]} \right) \quad [21]$$

$$R_a^{obs} = \lambda_2 \quad [22]$$

and R_a is a function of R_a^{obs} and other parameters which can be written as:

$$R_a = \frac{R_a^{obs}}{1 + \left(\left[\frac{RM_o^b}{R_a} \right] (R_b - R_a^{obs}) \right)} \approx \frac{R_a^{obs}}{1 + \left[\frac{M_o^b}{R_a} \right] (R_b - R_a^{obs})}, \text{ when } R \gg (R_b - R_a^{obs}) \quad [23]$$

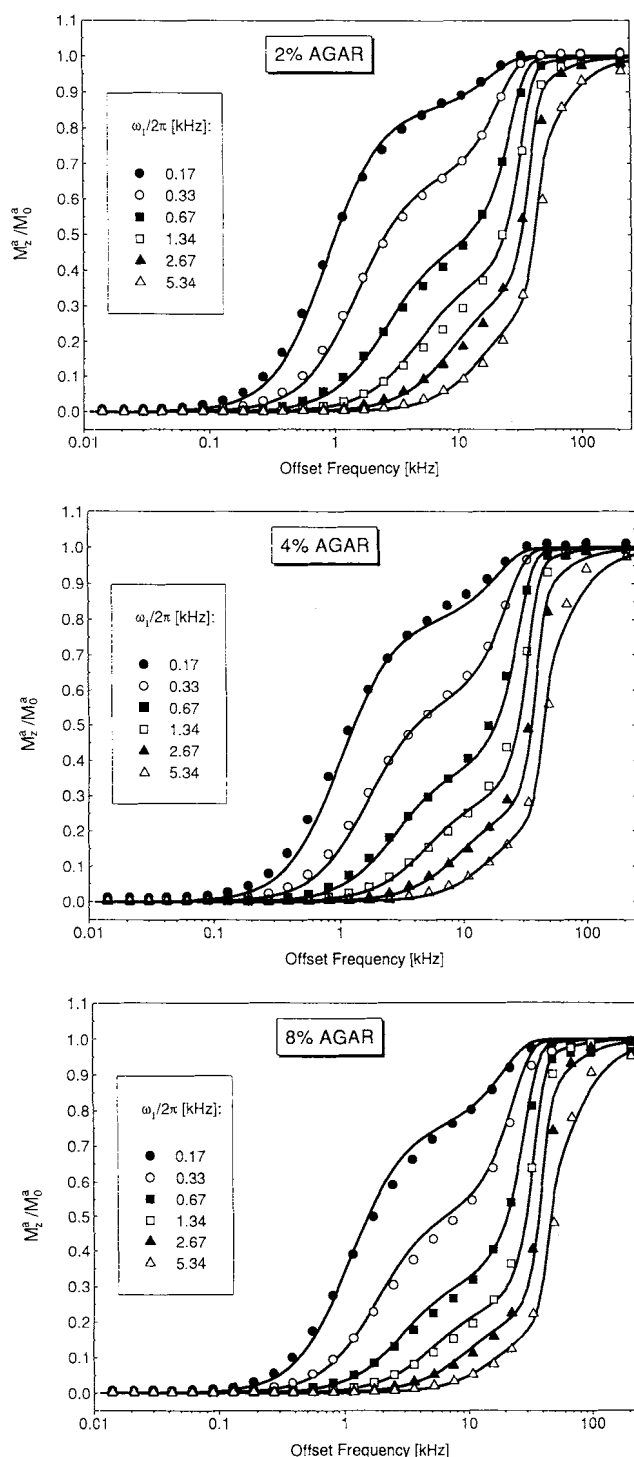


FIG. 7. (A) Magnetization transfer data for 2% agar. Solid lines result from a global fit to the complete set of 468 data points. T_1 and T_2 of pool A and the size of pool B are modelled as linear functions of the agar concentration as described in Eqs. [15–17]. The values of the fitted parameters are listed in Table 1. (B) Magnetization transfer data for 4% agar. (C) Magnetization transfer data for 8% agar.

Equation [23] can be seen intuitively to have the correct properties. If $R_b = R_a^{obs}$, then from the algebra in Eq. [23], R_a will be equal to R_a^{obs} which is expected when pool B has no effect. If R_b exceeds R , R_a will be less than R_a^{obs} . If

Table 1
Global Fit Parameters at 1.5 T

T_{2b}	$12.9 \pm 0.1 \mu\text{s}$
R_b	$1 \pm 4 \text{ s}^{-1}$
R	$176 \pm 22 \text{ s}^{-1}$
R_w/k	$1.22 \pm 0.07 \text{ s}^{-1}$
S_1/R_w	$0.32 \pm 0.03 [\% \text{ conc}]^{-1}$
S_2/R_w	$25.5 \pm 0.9 [\% \text{ conc}]^{-1}$

$R_b < R_o^{\text{obs}}$, the converse is true. We have measured the observed longitudinal relaxation rates for the 2%, 4%, and 8% agar samples with an inversion recovery experiment to be 0.49 ± 0.02 , 0.68 ± 0.03 , $1.14 \pm 0.05 \text{ s}^{-1}$, respectively. From these values, we can determine R_o for each of the gel concentrations. These, in turn, give us three similar estimates of R_w (0.30 , 0.30 , 0.32 s^{-1}) with an average of 0.31 s^{-1} . This information allows calculation of all physical parameters of the model at 1.5 T, as summarized in Table 2, and they can be evaluated to determine whether they are reasonable.

The complete experiment and equivalent analysis was also performed at 0.6 T to investigate the dependence of these magnetization transfer effects on field strength. Data which were quantitatively similar to those shown in Fig. 7, but quantitatively different at a significant level, were obtained (not shown). An independent analysis gave fits having an average residual deviation from the experimental points of 0.017. The relaxation rate for pure water obtained at 0.6 T was $0.27 \pm .02 \text{ s}^{-1}$, consistent with that obtained at 1.5 T.

The physical parameters of the model at 0.6 T are presented in Table 3. Again, they are a consistent and reasonable set of values. All the parameters associated with the magnetization transfer (RM_o^b , M_o^b , T_{2p} and R_b) are identical at 0.6 and 1.5 T within one standard deviation suggesting that the mechanism of magnetization transfer, at least in this system, is field independent. The major differences between 0.6 and 1.5 T are in the values of R_o , arising from the effect of the gel on water relaxation. S_1 in Eq. [15] is $0.128 \pm 0.004 \text{ s}^{-1} [\% \text{ conc}]^{-1}$ and $0.098 \pm 0.009 \text{ s}^{-1} [\% \text{ conc}]^{-1}$, respectively, at 0.6 and 1.5 T. Rotational immobilization of water at the gel surface appears to be a more effective relaxation mechanism at a Larmor frequency of 25 MHz than at 63 MHz.

DISCUSSION

The T_2 times for pool B are very short (12.9 and 12.7 μs) and are determined precisely ($\pm 0.1 \mu\text{s}$) by these fits. Val-

Table 2
Physical Parameters of the Model at 1.5 T Extracted from the Fit

	2% Agar	4% Agar	8% Agar
$R_a (\text{s}^{-1})$	0.51 ± 0.07	0.70 ± 0.10	1.08 ± 0.16
$T_{2a} (\text{ms})$	63 ± 8	32 ± 4	16 ± 2
$RM_o^b (\text{s}^{-1})$	0.9 ± 0.1	1.8 ± 0.2	3.9 ± 0.5
M_o^b	0.0051 ± 0.001	0.011 ± 0.002	0.022 ± 0.004
$T_{2b} (\mu\text{s})$		12.9 ± 0.1	
$R_b (\text{s}^{-1})$		1 ± 4	

Table 3
Physical Parameters of the Model at 0.6 T Extracted from the Fit

	2% Agar	4% Agar	8% Agar
$R_a (\text{s}^{-1})$	0.53 ± 0.07	0.80 ± 0.11	1.38 ± 0.19
$T_{2a} (\text{ms})$	75 ± 11	38 ± 16	18 ± 3
$RM_o^b (\text{s}^{-1})$	1.0 ± 0.2	2.1 ± 0.4	4.4 ± 0.7
M_o^b	0.0056 ± 0.001	0.012 ± 0.002	0.024 ± 0.003
$T_{2b} (\mu\text{s})$		12.7 ± 0.1	
$R_b (\text{s}^{-1})$		1 ± 5	

ues of T_{2b} outside this range fail to give statistically satisfactory fits to the experiment. Presumably, the stretched fibre structure of gel (21) leads to an exceptionally rigid macromolecular structure. The value obtained in this work is consistent with T_2 times of 9 μs obtained by Swanson for heat denatured egg white (29). It is significantly shorter than the 58 μs reported by Caines *et al.* (22) extracted from rat muscle measured by Grad *et al.* (19) or the 200 μs assumed by other authors (10, 23). Grad and Bryant (34) obtained a value of T_2 for crosslinked BSA of 10 μs . The longitudinal relaxation rate of the semi-solid phase cannot be determined from these experiments. This conclusion is in agreement with Edzes and Samulski (25).

For the 2% agar sample, the number of protons contributing to pool B is determined to be 0.005. From the chemical composition of agar (D-galactose + 3–6 anhydro-L-galactose chain; C_{12} , O_{11} , H_{18}) and with the knowledge that the powdered material was 85% agar and 15% water by weight, the number of protons in the 2% agar sample as a fraction of the number of protons in the water would be 0.0085. Given that some of the protons are inaccessible to the spin exchange process, and any labile protons that undergo rapid chemical exchange will not contribute, the estimate of M_o^b is very reasonable (21). The exchange rate from A to B in Table 2 (RM_o^b) increases linearly with agar concentration. Values of a few per second are generally consistent with the work of others (22, 29) but are quantitatively higher than the rate of 0.34 s^{-1} quoted by Eng *et al.* (2) for 1.5% agar. A reason for this discrepancy will be discussed later. The longitudinal relaxation rate for water at 1.5 T (R_w) is $0.31 \pm 0.04 \text{ s}^{-1}$, in good agreement with known values. Finally, the T_{2a} values extracted from the global fit at 1.5 and 0.6 T are compared in Table 4 with the values (T_2^{obs}) measured in CPMG experiments without saturation pulses. Agreement is fairly good considering the complete independence of the experiments.

Table 4
Comparison of Measured and Derived T_2 Values for the Liquid Pool at 1.5 and 0.6 T

	2% Agar	4% Agar	8% Agar
$T_{2a}^{\text{obs}} (\text{ms})$ 1.5 T	74 ± 4	40 ± 2	17 ± 1
$T_{2a} (\text{ms})$ 1.5 T	63 ± 8	32 ± 4	16 ± 2
$T_{2a}^{\text{obs}} (\text{ms})$ 0.6 T	66 ± 3	41 ± 2	21 ± 3
$T_{2a} (\text{ms})$ 0.6 T	75 ± 11	38 ± 6	18 ± 3

Thus, the model parameters extracted from the global fit of the magnetization transfer data for agar are reasonable and consistent with other values where they are known. This gives confidence that the fitting process is not just a mathematical exercise but is giving insight into the physical mechanisms.

One advantage of a quantitative, detailed and accurate model for magnetization transfer is that it allows aspects of the system which cannot be measured directly to be investigated as illustrated in Fig. 8. The solid line corresponding to pool A with an exchange rate from A to B of $RM_o^b = 0.9 \text{ s}^{-1}$ is the model result for 2% agar with RF amplitude of 0.67 kHz, identical to that shown in Fig. 7A. We can enquire what the magnetization in pool A would look like if there was no exchange (i.e., $RM_o^b = 0$). Under these conditions, pool A would be isolated and saturated by the off-resonance radiation through the direct effect. Thus, we see the familiar sigmoidal curve which corresponds to a Lorentzian line shape. At the other extreme, we could ask about the appearance of pool A if the rate of exchange were infinite (i.e., $RM_o^b = \infty$). When the exchange rate is infinitely large, the fraction of longitudinal spins (M_z^a/M_o^a in pool A or M_z^b/M_o^b in pool B) is the same in both pools. Therefore, we have plotted M_z^b/M_o^b as dashed lines on Fig. 8. The dashed and solid lines are superimposed for $RM_o^b = \infty$. It can be seen from Fig. 8 that the agar gel system is functioning well away from this limit with M_z^b/M_o^b only one half of M_z^a/M_o^a at 10 kHz offset (a limit which has been designated in the literature as a Z spectrum (34, 35). Finally, we can look at the direct irradiation effect on pool B when the exchange is zero. This curve simply reflects the Gaussian line shape of pool B plotted on a logarithmic plot of offset frequency.

Thus, we can determine from exploring nonmeasurable situations with the model that this particular gel experiment (2% agar with an H_1 amplitude of 0.67 kHz) is far from the limit of equivalent saturation of both pools

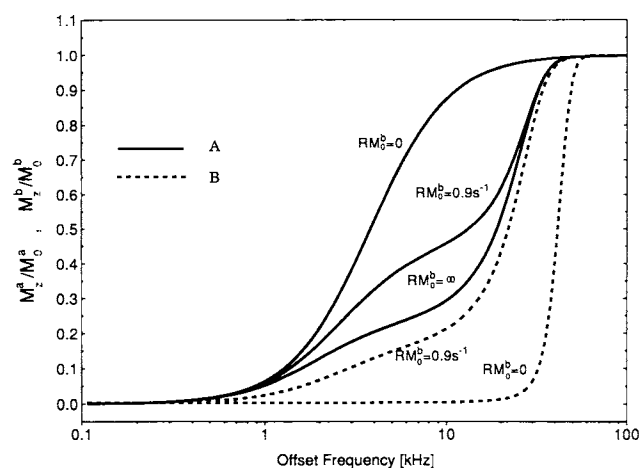


FIG. 8. Residual magnetization of pool A (solid) and pool B (dashed) for 2% agar and H_1 amplitude of 0.67 kHz. Both pools are shown for the fitted exchange rate from pool A to pool B of $RM_o^b = 0.9 \text{ s}^{-1}$ and for the predicted situation with no exchange $RM_o^b = 0$ and infinite by rapid exchange $RM_o^b = \infty$.

and far from the limit of pool B being fully saturated. By comparison, the analysis of MTC used by Balaban's group (1, 2, 36) which leads to the simple relationship:

$$RM_o^b = \frac{1}{T_{1sat}} [1 - M_z^a/M_o^a] \quad [24]$$

is predicated on the assumption that pool B is fully saturated. This assumption is clearly not valid for these agar systems and this assumption leads to the pseudo-first order rate constant for agar reported by Eng *et al.* (2) differing significantly from RM_o^b reported in Table 2.

Using the same model parameters and 2% agar, Fig. 9 shows graphically the extent of suppression achieved by magnetization transfer at 1.5 T for power levels 0.17, 0.67, and 2.67 kHz, compared with the direct effect on pool A alone. The direct effect shows the familiar sigmoidal curve translated by factors of 4 along the offset frequency axis. The shaded area (affectionately known in our lab as "the bite") is the additional saturation achieved due to magnetization transfer. Figure 9 shows that this type of off-resonance irradiation provides a limited window of opportunity for detecting magnetization transfer corresponding to rf amplitudes of about 1 kHz. As the RF amplitude is decreased, the MTC occurs at lower offset frequency but its magnitude is considerably diminished. Thus, it is not possible to reduce both RF amplitude and offset frequency to maintain contrast in the design of MTC imaging experiments (2). Similarly, at higher powers, the direct effect begins to crowd out the magnetization transfer leaving only a narrow band of offset frequencies having significant saturation of pool B with minimal direct effect on pool A. For example, with an H_1 amplitude of 0.67 kHz, it is possible to achieve an additional 30% saturation due to magnetization transfer beyond that produced by the direct effect over a band of offset

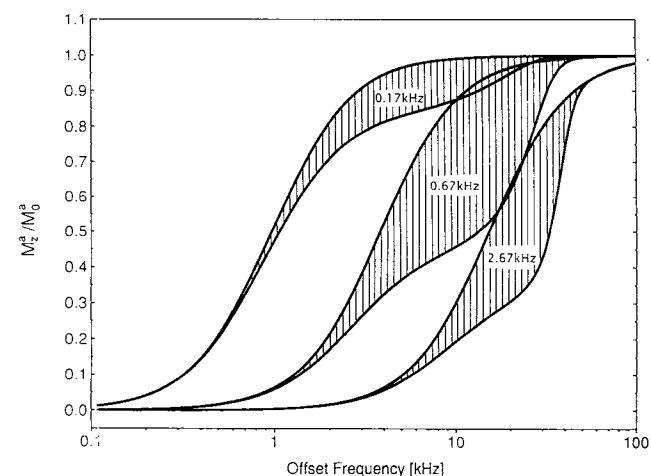


FIG. 9. The lower curve of each pair is the model fitted to the 2% gel data at H_1 amplitudes of 0.17, 0.67, and 2.67 kHz. The comparable direct effect with no exchange between the pools is shown as the upper curves. The shaded regions between the curves illustrate the saturation due to magnetization transfer.

frequencies with a range of 3.6 times. When the H_1 amplitude is increased to 2.67 kHz, however, the offset frequency band that can produce a 30% additional saturation has a frequency width of only 1.8 times. Figure 9, therefore, illustrates clearly the limitations of the off-resonance experiment for establishing MTC due to the competition between the direct effect and saturation of the semisolid pool. It is possible that selective pulsed irradiation schemes can saturate pool B more specifically without affecting pool A, but that is the subject of another paper.

CONCLUSIONS

We have presented a comprehensive set of magnetization transfer experiments in agar gels which is modelled accurately by simple exchange between two pools. The liquid pool has a Lorentzian line shape and the semisolid pool has a Gaussian line shape. The parameters extracted from fitting the model to the data are physically realistic. The parameters related to the exchange are the same at 0.6 and 1.5 T with the only difference being an increased relaxivity (1.3 times) of the liquid pool at the lower field, excluding the magnetization transfer effects. The model has then been used to show that neither assumptions of fast exchange between the pools nor complete saturation of the semisolid pool apply to these agar samples. Limitations of the simple off-resonance irradiation technique for MTC due to competition between direct effects and semisolid saturation are pointed out. This work serves as a theoretical framework for the rational design of MTC imaging experiments.

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REFERENCES

1. S. D. Wolff, R. S. Balaban, *Magn. Reson. Med.* **10**, 135 (1989).
2. J. Eng, T. L. Ceckler, R. S. Balaban, *Magn. Reson. Med.* **17**, 304 (1991).
3. S. D. Wolff, S. Chesnick, J. A. Frank, K. O. Lim, R. S. Balaban, *Radiology* **179**, 623 (1991).
4. E. Schneider, R. Prost, G. H. Glover, *J. Magn. Reson. Imaging*, **3**, 417 (1993).
5. E. Schneider, "Optimization of Pulsed Magnetization Transfer for Enhanced Synovial Fluid-Articular Cartilage Contrast," Radiological Society of N. America, (1992). [abstract #2754]
6. J. I. Tantt, R. E. Sepponen, R. J. Foust, E. Kinnunen, "MR of Multiple Sclerosis Plaques at 0.1 T: Correlation between Magnetization Transfer Contrast, $T_{1\rho}$ -dispersion, T_1 values and intensity in T_2 Weighted Images," Society for Magnetic Resonance in Medicine, 1990. [abstract]
7. R. J. Ordidge, J. A. Helpert, R. A. Knight, Z. Qing, K. M. A. Welch, *Magn. Reson. Imaging*, **9**, 895 (1991).
8. G. B. Pike, G. H. Glover, D. R. Enzmann, "Multisection Pulsed Magnetization Transfer Imaging of Multiple Sclerosis," Radiological Society of N. America, p. 208 (1991). [abstract]
9. G. B. Pike, B. S. Hu, G. H. Glover, D. R. Enzmann, *Magn. Reson. Med.* **25**, 372 (1992).
10. B. S. Hu, S. M. Conolly, G. A. Wright, D. G. Nishimura, A. Macovski, *Magn. Reson. Med.* **26**, 231 (1992).
11. H. N. Yeung, A. M. Aisen, *Radiology*, **183**, 209 (1992).
12. R. A. Jones, T. E. Southon, *J. Magn. Reson.* **97**, 171 (1992).
13. G. B. Pike, "Signal Behavior in Pulsed Magnetization Transfer Spoiled-GRASS" Radiological Society of N. America, (1992). [abstract #2133]
14. W. T. Dixon, H. Egels, M. Castillo, M. Sardashti, *Magn. Reson. Imaging* **8**, 417 (1990).
15. R. T. Constable, A. W. Anderson, J. Zhong, J. C. Gore, *Magn. Reson. Med.*, **28**, 9 (1992).
16. H. T. Edzes, E. T. Samulski, *J. Magn. Reson.* **31**, 207 (1978).
17. J. C. Gore, M. S. Brown, I. M. Armitage, *Magn. Reson. Med.* **9**, 333 (1989).
18. J. Zhong, J. C. Gore, "The Importance of Cross Relaxation to the Elevated Water T_1/T_2 Ratio below 2T," Society for Magnetic Resonance in Medicine, (1989). [abstract 212]
19. J. Grad, D. Mendelson, F. Hyder, R. G. Bryant, *J. Magn. Reson.* **86**, 416 (1990).
20. R. G. Bryant, D. A. Mendelson, C. C. Lester, *Magn. Reson. Med.* **21**, 117 (1991).
21. P. S. Belton, B. P. Hills, E. R. Raimbaud, *Mol. Phys.* **63**, 825 (1988).
22. G. H. Caines, T. Schleich, J. M. Rydzewski, *J. Magn. Reson.* **95**, 558 (1991).
23. X. Wu, *J. Magn. Reson.* **94**, 186 (1991).
24. S. Forsen, R. A. Hoffman, *J. Chem. Phys.* **45**, 2049 (1966).
25. H. T. Edzes, E. T. Samulski, *Nature* **265**, 521 (1977).
26. F. A. L. Anet, D. J. O'Leary, *J. Magn. Reson.* **86**, 358 (1990).
27. K. V. Vasavada, J. I. Kaplan, *J. Magn. Reson.* **62**, 37 (1985).
28. H. N. Yeung, S. D. Swanson, *J. Magn. Reson.* **99**, 466 (1992).
29. S. D. Swanson, in "Book of Abstracts, Annual Meeting, Society for Magnetic Resonance in Medicine, 1992," p. 255.
30. A. Abragam, in "The Principles of Nuclear Magnetism," Oxford Press, London, 1961.
31. J. I. Kaplan, K. V. Vasavada, *J. Magn. Reson.* **52**, 475 (1983).
32. W. H. Press, B. P. Flannery, S. A. Teukolsky, W. T. Vetterling, in "Numerical Recipes," Cambridge University Press, Cambridge, 1986.
33. S. Forsen, R. A. Hoffman, *J. Chem. Phys.* **39**, 2892 (1963).
34. J. Grad, R. G. Bryant, *J. Magn. Reson.* **90**, 1 (1990).
35. D. A. Mendelson, J. F. Heinsbergen, S. D. Kennedy, L. S. Szczepaniak, C. C. Lester, R. G. Bryant, *Magn. Reson. Imaging* **9**, 975 (1991).
36. S. D. Wolff, R. S. Balaban, *J. Magn. Reson.* **86**, 164 (1990).